

# LAGRANGE MULTIPLIER

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## ■ EXTREMUM PROBLEMS WITH CONSTRAINTS

In Example 6 of the last section, we solved the problem of minimizing

$$S = xy + 2xz + 2yz \quad (1)$$

subject to the constraint

$$xyz - 32 = 0 \quad (2)$$

This is a special case of the following general problem:

### **13.9.1** *Three-Variable Extremum Problem with One Constraint*

Maximize or minimize the function  $f(x, y, z)$  subject to the constraint  $g(x, y, z) = 0$ .

We will also be interested in the following two-variable version of this problem:

### **13.9.2** *Two-Variable Extremum Problem with One Constraint*

Maximize or minimize the function  $f(x, y)$  subject to the constraint  $g(x, y) = 0$ .

# What is Lagrange Multiplier Theory?

- The Lagrange multiplier technique lets you find the maximum or minimum of a multivariable function  $f(x, y, \dots)$  when there is some constraint on the input values you are allowed to use.

Suppose you want to maximize this function:

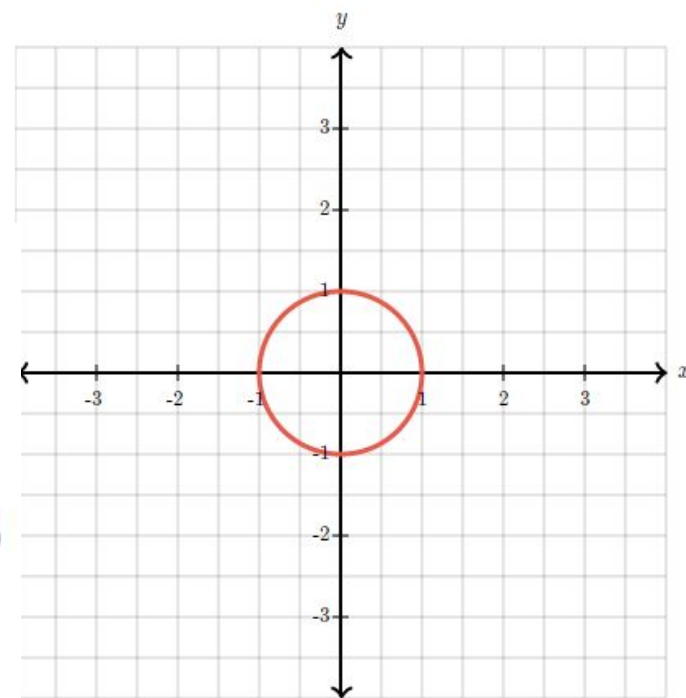
$$f(x, y) = 2x + y$$

But let's also say you limited yourself to inputs  $(x, y)$  which satisfy the following equation:

$$x^2 + y^2 = 1$$

In other words, for which point  $(x, y)$  on the **unit circle** is the value  $2x + y$  biggest?

This is what's known as a **constrained optimization problem**. The restriction to points where  $x^2 + y^2 = 1$  is called a "constraint", and  $f(x, y) = 2x + y$  is the function that needs to be optimized.



points  $(x, y)$  satisfying  $x^2 + y^2 = 1$ , also known as the unit circle.

$$\mathcal{L}(x, \lambda) = f(x) - \lambda g(x)$$

$\mathcal{L}$  = Lagrangian

$\lambda$  = Lagrange multiplier

$g(x)$  = equality constraint

$f(x)$  = function

$x$  = integer



## ■ LAGRANGE MULTIPLIERS

One way to attack problems of these types is to solve the constraint equation for one of the variables in terms of the others and substitute the result into  $f$ . This produces a new function of one or two variables that incorporates the constraint and can be maximized or minimized by applying standard methods. For example, to solve the problem in Example 6 of the last section we substituted (2) into (1) to obtain

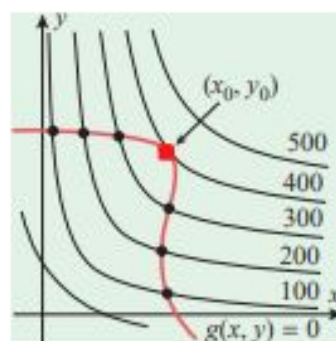
$$S = xy + \frac{64}{y} + \frac{64}{x}$$

which we then minimized by finding the critical points and applying the second partials test. However, this approach hinges on our ability to solve the constraint equation for one of the variables in terms of the others. If this cannot be done, then other methods must be used. One such method, called the *method of Lagrange multipliers*, will be discussed in this section.

To motivate the method of Lagrange multipliers, suppose that we are trying to maximize a function  $f(x, y)$  subject to the constraint  $g(x, y) = 0$ . Geometrically, this means that we are looking for a point  $(x_0, y_0)$  on the graph of the constraint curve at which  $f(x, y)$  is as large as possible. To help locate such a point, let us construct a contour plot of  $f(x, y)$  in the same coordinate system as the graph of  $g(x, y) = 0$ . For example, Figure 13.9.1a shows some typical level curves of  $f(x, y) = c$ , which we have labeled  $c = 100, 200, 300, 400$ , and 500 for purposes of illustration. In this figure, each point of intersection of  $g(x, y) = 0$  with a level curve is a candidate for a solution, since these points lie on the constraint curve. Among the seven such intersections shown in the figure, the maximum value of  $f(x, y)$  occurs at the intersection  $(x_0, y_0)$  where  $f(x, y)$  has a value of 400. Note that at  $(x_0, y_0)$  the constraint curve and the level curve just touch and thus have a *common* tangent line at this point. Since  $\nabla f(x_0, y_0)$  is normal to the level curve  $f(x, y) = 400$  at  $(x_0, y_0)$ , and since  $\nabla g(x_0, y_0)$  is normal to the constraint curve  $g(x, y) = 0$  at  $(x_0, y_0)$ , we conclude that the vectors  $\nabla f(x_0, y_0)$  and  $\nabla g(x_0, y_0)$  must be parallel. That is,

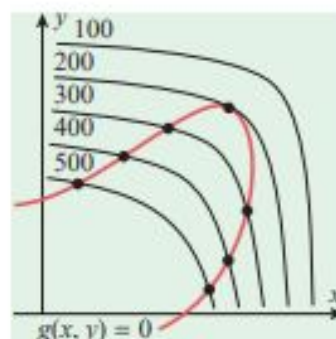
$$\nabla f(x_0, y_0) = \lambda \nabla g(x_0, y_0) \quad (3)$$

for some scalar  $\lambda$ . The same condition holds at points on the constraint curve where  $f(x, y)$  has a minimum. For example, if the level curves are as shown in Figure 13.9.1b, then the minimum value of  $f(x, y)$  occurs where the constraint curve just touches a level curve.



Maximum of  $f(x, y)$  is 400

(a)

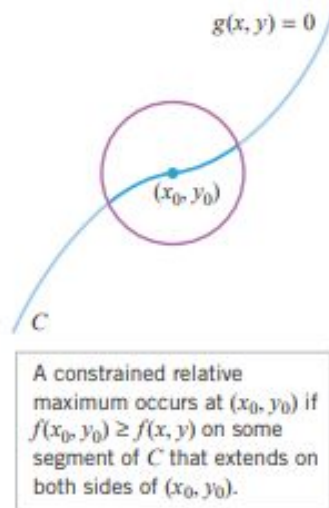


Minimum of  $f(x, y)$  is 200

(b)

▲ Figure 13.9.1





▲ Figure 13.9.2

Thus, to find the maximum or minimum of  $f(x, y)$  subject to the constraint  $g(x, y) = 0$ , we look for points at which (3) holds—this is the method of Lagrange multipliers.

Our next objective in this section is to make the preceding intuitive argument more precise. For this purpose it will help to begin with some terminology about the problem of maximizing or minimizing a function  $f(x, y)$  subject to a constraint  $g(x, y) = 0$ . As with other kinds of maximization and minimization problems, we need to distinguish between relative and absolute extrema. We will say that  $f$  has a **constrained absolute maximum (minimum)** at  $(x_0, y_0)$  if  $f(x_0, y_0)$  is the largest (smallest) value of  $f$  on the constraint curve, and we will say that  $f$  has a **constrained relative maximum (minimum)** at  $(x_0, y_0)$  if  $f(x_0, y_0)$  is the largest (smallest) value of  $f$  on some segment of the constraint curve that extends on both sides of the point  $(x_0, y_0)$  (Figure 13.9.2).

Let us assume that a constrained relative maximum or minimum occurs at the point  $(x_0, y_0)$ , and for simplicity let us further assume that the equation  $g(x, y) = 0$  can be smoothly parametrized as

$$x = x(s), \quad y = y(s)$$

where  $s$  is an arc length parameter with reference point  $(x_0, y_0)$  at  $s = 0$ . Thus, the quantity

$$z = f(x(s), y(s))$$

has a relative maximum or minimum at  $s = 0$ , and this implies that  $dz/ds = 0$  at that point. From the chain rule, this equation can be expressed as

$$\frac{dz}{ds} = \frac{\partial f}{\partial x} \frac{dx}{ds} + \frac{\partial f}{\partial y} \frac{dy}{ds} = \left( \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} \right) \cdot \left( \frac{dx}{ds} \mathbf{i} + \frac{dy}{ds} \mathbf{j} \right) = 0$$

where the derivatives are all evaluated at  $s = 0$ . However, the first factor in the dot product is the gradient of  $f$ , and the second factor is the unit tangent vector to the constraint curve. Since the point  $(x_0, y_0)$  corresponds to  $s = 0$ , it follows from this equation that

$$\nabla f(x_0, y_0) \cdot \mathbf{T}(0) = 0$$

which implies that the gradient is either  $\mathbf{0}$  or is normal to the constraint curve at a constrained relative extremum. However, the constraint curve  $g(x, y) = 0$  is a level curve for the function  $g(x, y)$ , so that if  $\nabla g(x_0, y_0) \neq \mathbf{0}$ , then  $\nabla g(x_0, y_0)$  is normal to this curve at  $(x_0, y_0)$ . It then follows that there is some scalar  $\lambda$  such that

$$\nabla f(x_0, y_0) = \lambda \nabla g(x_0, y_0) \quad (4)$$

This scalar is called a **Lagrange multiplier**. Thus, the **method of Lagrange multipliers** for finding constrained relative extrema is to look for points on the constraint curve  $g(x, y) = 0$  at which Equation (4) is satisfied for some scalar  $\lambda$ .



## Lagrange Multiplier (3.9)

A function  $(f(x, y, z))$  and a constraint function  $(g(x, y, z))$  will be given. Find  $f_x, f_y, f_z$  &  $g_x, g_y, g_z$ . Then,

$$f_x = \lambda g_x \quad \lambda = \text{Lagrange multiplier}$$

$$f_y = \lambda g_y$$

$$f_z = \lambda g_z$$

Solve the 3 equations to find the

value of  $x, y, z$  &  $\lambda$ .

\* The constraint function is another equation that is used with the above 3.

Plug in the values of  $x, y, z$  in the main function to find maximum & minimum value.



$$\nabla f(x_0, y_0) = \lambda \nabla g(x_0, y_0)$$

Get  $\nabla f$

Get  $\nabla g$

Equate  $\nabla f = \lambda \nabla g \dots\dots$

$$\begin{aligned}\nabla f &= f_x \hat{i} + f_y \hat{j} \\ &= \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j}\end{aligned}$$

$$\begin{aligned}\nabla g &= g_x \hat{i} + g_y \hat{j} \\ &= \frac{\partial g}{\partial x} \hat{i} + \frac{\partial g}{\partial y} \hat{j}\end{aligned}$$

$$\begin{aligned}\nabla f &= \lambda \nabla g \\ \Rightarrow f_x \hat{i} + f_y \hat{j} &= \lambda (g_x \hat{i} + g_y \hat{j})\end{aligned}$$

$$\Rightarrow f_x = \lambda g_x \quad \text{And} \quad f_y = \lambda g_y$$

# **PRACTICE PROBLEMS**

► **Example 1** At what point or points on the circle  $x^2 + y^2 = 1$  does  $f(x, y) = xy$  have an absolute maximum, and what is that maximum?

**Solution.** The circle  $x^2 + y^2 = 1$  is a closed and bounded set and  $f(x, y) = xy$  is a continuous function, so it follows from the Extreme-Value Theorem (Theorem 13.8.3) that  $f$  has an absolute maximum and an absolute minimum on the circle. To find these extrema, we will use Lagrange multipliers to find the constrained relative extrema, and then we will evaluate  $f$  at those relative extrema to find the absolute extrema.

We want to maximize  $f(x, y) = xy$  subject to the constraint

$$g(x, y) = x^2 + y^2 - 1 = 0 \quad (5)$$

First we will look for constrained *relative* extrema. For this purpose we will need the gradients

$$\nabla f = y\mathbf{i} + x\mathbf{j} \quad \text{and} \quad \nabla g = 2x\mathbf{i} + 2y\mathbf{j}$$

From the formula for  $\nabla g$  we see that  $\nabla g = \mathbf{0}$  if and only if  $x = 0$  and  $y = 0$ , so  $\nabla g \neq \mathbf{0}$  at any point on the circle  $x^2 + y^2 = 1$ . Thus, at a constrained relative extremum we must have

$$\nabla f = \lambda \nabla g \quad \text{or} \quad y\mathbf{i} + x\mathbf{j} = \lambda(2x\mathbf{i} + 2y\mathbf{j})$$

which is equivalent to the pair of equations

$$y = 2x\lambda \quad \text{and} \quad x = 2y\lambda$$

It follows from these equations that if  $x = 0$ , then  $y = 0$ , and if  $y = 0$ , then  $x = 0$ . In either case we have  $x^2 + y^2 = 0$ , so the constraint equation  $x^2 + y^2 = 1$  is not satisfied. Thus, we can assume that  $x$  and  $y$  are nonzero, and we can rewrite the equations as

$$\lambda = \frac{y}{2x} \quad \text{and} \quad \lambda = \frac{x}{2y}$$

from which we obtain

$$\frac{y}{2x} = \frac{x}{2y}$$

or

$$y^2 = x^2 \quad (6)$$

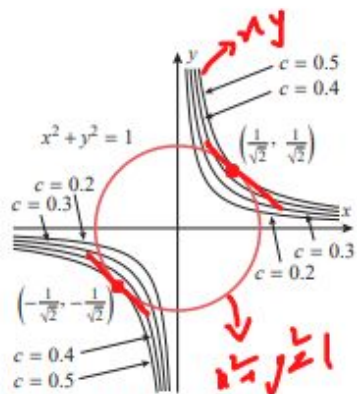
Substituting this in (5) yields

$$2x^2 - 1 = 0$$

from which we obtain  $x = \pm 1/\sqrt{2}$ . Each of these values, when substituted in Equation (6), produces  $y$ -values of  $y = \pm 1/\sqrt{2}$ . Thus, constrained relative extrema occur at the points  $(1/\sqrt{2}, 1/\sqrt{2})$ ,  $(1/\sqrt{2}, -1/\sqrt{2})$ ,  $(-1/\sqrt{2}, 1/\sqrt{2})$ , and  $(-1/\sqrt{2}, -1/\sqrt{2})$ . The values of  $xy$  at these points are as follows:

$$\Delta f = \lambda \nabla g$$

| $(x, y)$ | $(1/\sqrt{2}, 1/\sqrt{2})$ | $(1/\sqrt{2}, -1/\sqrt{2})$ | $(-1/\sqrt{2}, 1/\sqrt{2})$ | $(-1/\sqrt{2}, -1/\sqrt{2})$ |
|----------|----------------------------|-----------------------------|-----------------------------|------------------------------|
| $xy$     | $1/2$                      | $-1/2$                      | $-1/2$                      | $1/2$                        |



▲ Figure 13.9.3

Thus, the function  $f(x, y) = xy$  has an absolute maximum of  $\frac{1}{2}$  occurring at the two points  $(1/\sqrt{2}, 1/\sqrt{2})$  and  $(-1/\sqrt{2}, -1/\sqrt{2})$ . Although it was not asked for, we can also see that  $f$  has an absolute minimum of  $-\frac{1}{2}$  occurring at the points  $(1/\sqrt{2}, -1/\sqrt{2})$  and  $(-1/\sqrt{2}, 1/\sqrt{2})$ . Figure 13.9.3 shows some level curves  $xy = c$  and the constraint curve in the vicinity of the maxima. A similar figure for the minima can be obtained using negative values of  $c$  for the level curves  $xy = c$ . ◀



Find the point on the line  $y=2x+3$  that is closest to  $(4,2)$ .

$$D^2 = (x-4)^2 + (y-2)^2$$

$$f(x,y) = (x-4)^2 + (y-2)^2$$

$$g(x,y) = y - 2x - 3$$

$$\nabla f(x,y) = \left( \frac{\partial f}{\partial x} i + \frac{\partial f}{\partial y} j \right) [(x-4)^2 + (y-2)^2]$$
$$= 2(x-4) \hat{i} + 2(y-2) \hat{j}$$

$$\nabla g(x,y) = \left( \frac{\partial g}{\partial x} \hat{i} + \frac{\partial g}{\partial y} \hat{j} \right) (y - 2x - 3)$$
$$= -2\hat{i} + \hat{j}$$

$$\nabla f = \lambda \nabla g$$
$$2(x-4)\hat{i} + 2(y-2)\hat{j} = \lambda(-2\hat{i} + \hat{j})$$

comparing the coefficients

$$\begin{array}{l|l} 2(x-4) = -2\lambda & 2(y-2) = \lambda \\ \lambda = -(x-4) & \end{array}$$

$$\therefore -(x-4) = 2(y-2)$$

$$-x+4 = 2y-4$$
$$\Rightarrow y = \frac{1}{2}(-x+8)$$

Substituting this value in constraint eqn

$$\frac{1}{2}(-x+8) = 2x+3 \quad [y=2x+3]$$

$$\Rightarrow -x+8 = 4x+6$$

$$\Rightarrow 5x = 2$$

$$x = \frac{2}{5}$$

$$\therefore y = 2\left(\frac{2}{5}\right) + 3 = \frac{19}{5}$$

Find the point on the line  $y = 2x + 3$   
that is closest to  $(4, 2)$ .

$$f(x,y) = 4x^3 + y^2 \quad g(x,y) = 2x^2 + y^2 - 1$$

$$\therefore \text{let } g(x,y) = 2x^2 + y^2 - 1 = 0$$

$$\text{We know, } \vec{\nabla} f = \lambda \vec{\nabla} g$$

$$\left( \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} \right) (4x^3 + y^2) = \lambda \left[ \left( \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} \right) (2x^2 + y^2 - 1) \right]$$

$$\Rightarrow 12x^2 \hat{i} + 2y \hat{j} = 4x \lambda \hat{i} + 2y \lambda \hat{j}$$

Comparing the coefficients

$$2y = 2y \lambda$$

$$\Rightarrow \lambda = 1$$

$$\text{Again, } 12x^2 = 4x \lambda$$

putting the value of  $\lambda$

$$\Rightarrow 12x^2 - 4x = 0$$

$$\Rightarrow 4x(3x - 1) = 0$$

$$\therefore \text{either } 4x = 0 \text{ or } 3x - 1 = 0$$

$$\Rightarrow x = 0, \frac{1}{3}$$

$$\text{Again, putting } x = \frac{1}{3} \text{ in } 2x^2 + y^2 = 1$$

$$\Rightarrow 2\left(\frac{1}{3}\right)^2 + y^2 = 1$$

$$y = \pm \sqrt{7/9}$$

$$y = \pm \sqrt{7}/3$$

$$\left(\frac{1}{3}, \sqrt{7}/3\right) \left(\frac{1}{3}, -\sqrt{7}/3\right)$$

$$\text{when } x = 0,$$

$$2(0)^2 + y^2 = 1$$

$$\Rightarrow y = \pm 1$$

$$(0, 1) (0, -1)$$

$$\text{Now, } f(x,y) = 4x^3 + y^2$$

$$f\left(\frac{1}{3}, \sqrt{7}/3\right) = 25/27 \quad \left. \begin{array}{l} f\left(\frac{1}{3}, -\sqrt{7}/3\right) = 25/27 \end{array} \right\} \text{minimum}$$

$$f(0, 1) = 1 \quad \left. \begin{array}{l} f(0, -1) = 1 \end{array} \right\} \text{maximum}$$

Minimum value of  $f(x,y)$  is  $25/27$  it occurs at  $\left(\frac{1}{3}, \pm \sqrt{7}/3\right)$

Maximum value of  $f(x,y)$  is  $1$  it occurs at  $(0, \pm 1)$



$$f(x,y) = x^2 - y^2, \quad x^2 + y^2 = 25$$

$$\text{let } g(x,y) = x^2 + y^2 - 25 = 0$$

$$\vec{\nabla} f = \lambda \cdot \vec{\nabla} g$$

$$\therefore \left( \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} \right) (x^2 - y^2) = \lambda \left[ \left( \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} \right) (x^2 + y^2 - 25) \right]$$

$$\Rightarrow 2x \hat{i} - 2y \hat{j} = \lambda [2x \hat{i} + 2y \hat{j}]$$

Comparing the coefficients of both sides

$$\begin{array}{l|l} 2x = 2\lambda x & -2y = -2y\lambda \\ \Rightarrow \lambda = 1 & \Rightarrow \lambda = -1 \end{array}$$

$$x \neq 0, y = 0 \quad \therefore \lambda = 1$$

$$x^2 + y^2 = 25$$

$$\text{let } y = 0$$

$$x^2 = 25$$

$$\therefore x = \pm 5$$

$$(5, 0) \quad (-5, 0)$$

$$y \neq 0, x = 0, \lambda = -1$$

$$x^2 + y^2 = 25$$

$$y^2 = 25$$

$$\Rightarrow y = \pm 5$$

$$(0, 5) \quad (0, -5)$$

|                       | max       | min       | max      | min      |
|-----------------------|-----------|-----------|----------|----------|
| $(x, y)$              | $(-5, 0)$ | $(0, -5)$ | $(5, 0)$ | $(0, 5)$ |
| $f(x, y) = x^2 - y^2$ | 25        | -25       | 25       | -25      |

Maximum value of  $f(x, y)$  is 25 it occurs at  $(\pm 5, 0)$

Minimum value of  $f(x, y)$  is -25 it occurs at  $(0, \pm 5)$

Find the maximum and minimum values of  $f(x,y) = 3x+4y$  subject to  $x^2+y^2=100$

Objective function

constraint function

$$\therefore \text{Let } g(x,y) = x^2+y^2-100=0$$

$$\therefore \nabla f = \lambda \cdot \nabla g$$

$$\therefore f_x = \lambda g_x$$

$$f_y = \lambda g_y$$

$$f_x = 3, f_y = 4, g_x = 2x, g_y = 2y$$

$$\therefore f_x = \lambda g_x$$

$$3 = \lambda \times 2x$$

$$\Rightarrow x = \frac{3}{2\lambda}$$

$$\therefore f_y = \lambda g_y$$

$$4 = \lambda \times 2y$$

$$y = \frac{4}{2\lambda} = \frac{2}{\lambda}$$

$$\text{Again, } x^2+y^2-100=0$$

putting the value of  $x$  &  $y$

$$\text{Again, } x^2+y^2-100=0$$

putting the value of  $x$  &  $y$

$$\therefore \left(\frac{3}{2\lambda}\right)^2 + \left(\frac{2}{\lambda}\right)^2 - 100 = 0$$

$$\Rightarrow \frac{9}{4\lambda^2} + \frac{4}{\lambda^2} - 100 = 0$$

$$\Rightarrow \frac{9+16}{4\lambda^2} = 100$$

$$\Rightarrow 25 = 400\lambda^2$$

$$\Rightarrow \lambda = \pm \frac{1}{4}$$

$$\therefore x = \frac{3}{2\lambda}, y = \frac{2}{\lambda}$$

$$\text{when } \lambda = \frac{1}{4}$$

$$x = 6, y = 8$$

$$\text{when } \lambda = -\frac{1}{4}$$

$$x = -6, y = -8$$

$$\therefore (6,8) \quad (-6,-8)$$

$$\text{Now, } f(x,y) = 3x+4y$$

$$\text{At } (6,8) \quad f(6,8) = 50$$

$$\text{At } (-6,-8) \quad f(-6,-8) = -50$$

Maximum value of  $f(x,y)$  is 50 it occurs at  $(6,8)$

Minimum value of  $f(x,y)$  is -50 it occurs at  $(-6,-8)$



Find the maximum and minimum values of  $f(x,y) = y^2 - 4x^2$   
subject to  $x^2 + 2y^2 = 4$

$$\text{Let } g(x,y) = x^2 + 2y^2 - 4 = 0$$

$$\nabla f = \lambda \nabla g$$

$$f_x = -8x, \quad g_x = 2x, \quad f_y = 2y, \quad g_y = 4y.$$

$$f_x = \lambda g_x$$

$$\therefore -8x = \lambda \times 2x$$

$$2\lambda x + 8x = 0$$

$$\Rightarrow 2x(\lambda + 4) = 0$$

either,

$$2x = 0$$

$$\therefore x = 0$$

$$\text{or } \lambda + 4 = 0$$

$$\Rightarrow \lambda = -4$$

$$f_y = \lambda g_y$$

$$2y = \lambda \times 4y$$

$$4\lambda y - 2y = 0$$

$$\Rightarrow 2y(2\lambda - 1) = 0$$

either  $2y = 0$

$$\Rightarrow y = 0$$

$$\text{or } 2\lambda - 1 = 0$$

$$\Rightarrow \lambda = 1/2.$$

Using the constraint function:  $x^2 + 2y^2 - 4 = 0$



when  $x=0$ ,

$$0^2 + 2y^2 - 4 = 0$$

$$\Rightarrow y = \pm \sqrt{2}$$

$$(0, \sqrt{2}) \quad (0, -\sqrt{2})$$

when  $y=0$ ,

$$x^2 + 2(0)^2 - 4 = 0$$

$$x = \pm 2$$

$$(2, 0) \quad (-2, 0)$$

$$f(x, y) = y^2 - 4x^2$$

$$f(0, \sqrt{2}) = (\sqrt{2})^2 - 4(0)^2 = 2 \quad \left. \vphantom{f(0, \sqrt{2})} \right\} \text{maximum}$$

$$f(0, -\sqrt{2}) = (-\sqrt{2})^2 - 4(0)^2 = 2$$

$$f(2, 0) = (0)^2 - 4(2)^2 = -16 \quad \left. \vphantom{f(2, 0)} \right\} \text{minimum}$$

$$f(\cancel{2, 0}) = (0)^2 - 4(-2)^2 = -16$$

$(-2, 0)$

Maximum value of  $f(x, y)$  is 2 it occurs at  $(0, \pm\sqrt{2})$

Minimum value of  $f(x, y)$  is -16 it occurs at  $(\pm 2, 0)$

$$\begin{aligned}
 \underline{10.} \quad f(x, y, z) &= 3x + 6y + 2z ; \quad 2x^2 + 4y^2 + z^2 = 70 \\
 f_x &= 3 ; f_y = 6 ; f_z = 2 \quad \Rightarrow \quad 2x^2 + 4y^2 + z^2 - 70 = 0 \quad \text{--- ①} \\
 g_x &= 4x ; g_y = 8y ; g_z = 2z \\
 \therefore 3 &= \lambda 4x \quad 6 = \lambda \times 8y \quad 2 = \lambda \times 2z \\
 \Rightarrow x\lambda &= \frac{3}{4\lambda} \quad \Rightarrow y = \frac{6}{8\lambda} \quad \Rightarrow z = \frac{2}{2\lambda}
 \end{aligned}$$

$$\therefore \text{①} \Rightarrow 2\left(\frac{3}{4\lambda}\right)^2 + 4\left(\frac{6}{8\lambda}\right)^2 + \left(\frac{2}{2\lambda}\right)^2 - 70 = 0$$

$$\Rightarrow \frac{9}{8\lambda^2} + \frac{9}{4\lambda^2} + \frac{1}{\lambda^2} - 70 = 0$$

$$\Rightarrow \frac{9 + 18 + 8 - 560\lambda^2}{8\lambda^2} = 0$$

$$\Rightarrow 35 - 560\lambda^2 = 0$$

$$\Rightarrow 1 - 16\lambda^2 = 0$$

$$\Rightarrow \lambda = \pm \frac{1}{4}$$

$$\text{when } \lambda = 1/4$$

$$x = -3 ; y = -3 ; z = -4$$

$$(-3, -3, -4)$$

$$\text{when } \lambda = 1/4$$

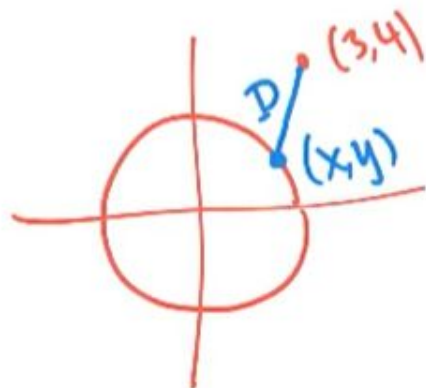
$$x = 3 ; y = 3 ; z = 4$$

$$(3, 3, 4)$$

10.  $f(x, y, z) = 3x + 6y + 2z$  ;  $2x^2 + 4y^2 + 2z^2 = 1$



3. Find the point on the circle  $x^2 + y^2 = 4$  that is closest to the point  $(3, 4)$ .



$$D = \sqrt{(x-3)^2 + (y-4)^2}$$

$$f(x,y) = (x-3)^2 + (y-4)^2$$

$$g(x,y) = x^2 + y^2 - 4 = 0$$

$$f_x = \lambda g_x \rightarrow 2(x-3) = \lambda \cdot 2x \rightarrow 2x-6 = 2x\lambda$$

$$f_y = \lambda g_y \rightarrow 2(y-4) = \lambda \cdot 2y$$

$$2x - 2x\lambda = 6$$

$$x(2-2\lambda) = 6$$

$$x = \frac{6}{2-2\lambda} = \frac{3}{1-\lambda}$$

$$2y - 8 = 2y\lambda$$

$$2y - 2y\lambda = 8$$

$$y(2-2\lambda) = 8$$

$$y = \frac{4}{1-\lambda}$$

$$\left(\frac{3}{1-\lambda}\right)^2 + \left(\frac{4}{1-\lambda}\right)^2 = 4$$

$$\frac{9}{(1-\lambda)^2} + \frac{16}{(1-\lambda)^2} = 4$$

$$\frac{9}{(1-\lambda)^2} + \frac{16}{(1-\lambda)^2} = 4$$

$$x = \frac{3}{1-\lambda}$$

$$y = \frac{4}{1-\lambda}$$

$$\lambda = -\frac{3}{2}: x = \frac{6}{5}, y = \frac{8}{5}$$

$$\lambda = \frac{7}{2}: x = -\frac{6}{5}, y = -\frac{8}{5}$$

$$\frac{25}{(1-\lambda)^2} = 4$$

$$25 = 4(1-\lambda)^2$$

$$\frac{25}{4} = (1-\lambda)^2$$

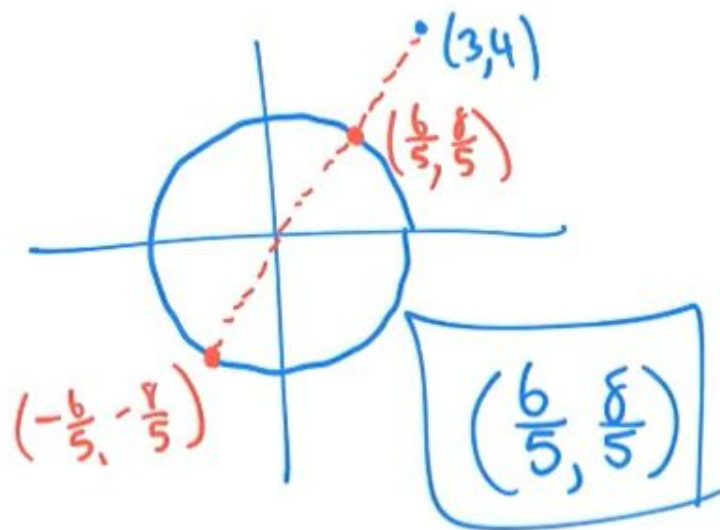
$$1-\lambda = \pm \frac{5}{2}$$

$$1-\lambda = \frac{5}{2}$$

$$\lambda = -\frac{3}{2}$$

$$1-\lambda = -\frac{5}{2}$$

$$\lambda = \frac{7}{2}$$



4. In economics, a Cobb-Douglas equation is used to compute total production  $Y$  as a function of labor input  $L$  and capital input  $K$ . One such equation is  $Y = 32L^{0.6}K^{0.4}$ . Maximize production subject to the constraint  $4L + 2K = 50$ .

$$f(L, K) = 32L^{0.6}K^{0.4}$$

$$g(L, K) = 4L + 2K - 50$$

$$f_L = \lambda g_L \rightarrow 32 \cdot K^{0.4} \cdot 0.6L^{-0.4} = \lambda \cdot 4$$

$$f_K = \lambda g_K$$

$$\downarrow$$

$$32 \cdot L^{0.6} \cdot (0.4)K^{-0.6} = \lambda \cdot 2$$

$$6.4 \frac{L^{0.6}}{K^{0.6}} = \lambda$$

$$\left( \cancel{K^{0.6}} \cancel{L^{0.4}} \left( 19.2 \cdot \frac{K^{0.4}}{\cancel{L^{0.4}}} \right) = 4 \left( 6.4 \frac{\cancel{L^{0.6}}}{\cancel{K^{0.6}}} \right) \right) \left( \cancel{L^{0.4}} \cancel{K^{0.6}} \right)$$

$$19.2K = 25.6L$$

$$K = \frac{4}{3}L$$

See the working in the next slide



①

$$f_K = \lambda g_K$$

$$32 L^{0.6} (0.4) K^{-0.6} = \lambda \cdot 2$$

$$\Rightarrow 12.8 \frac{L^{0.6}}{K^{0.6}} = \lambda \cdot 2$$

$$\Rightarrow \lambda = 6.4 \frac{L^{0.6}}{K^{0.6}}$$

②

$$f_L = \lambda g_L$$

$$32 K^{0.4} (0.6) L^{-0.4} = \lambda \cdot 4$$

$$\Rightarrow 19.2 \frac{K^{0.4}}{L^{0.4}} = \left( \frac{6.4 L^{0.6}}{K^{0.6}} \right) \cdot 4$$

→ Multiply both sides with  $K^{0.6} L^{0.4}$ ,

$$\cancel{K^{0.6}} \cancel{L^{0.4}} \left( 19.2 \frac{\cancel{K^{0.4}}}{\cancel{L^{0.4}}} \right) = \left( 6.4 \frac{\cancel{L^{0.6}}}{\cancel{K^{0.6}}} \right) 4 \cancel{K^{0.6}} \cancel{L^{0.4}}$$

$$\Rightarrow 19.2 K = 25.6 L$$

$$\Rightarrow K = \frac{4}{3} L$$

$$4L + 2K = 50$$

$$K = \frac{4}{3}L$$

$$4L + 2\left(\frac{4}{3}L\right) = 50$$

$$4L + \frac{8}{3}L = 50$$

$$\frac{12}{3}L + \frac{8}{3}L = 50$$

$$\frac{20}{3}L = 50$$

$$L = 50 \cdot \frac{3}{20}$$

$$= 7.5$$

$$K = \frac{4}{3}(7.5)$$

$$= 10$$

Max is at

$$L = 7.5$$

$$K = 10$$